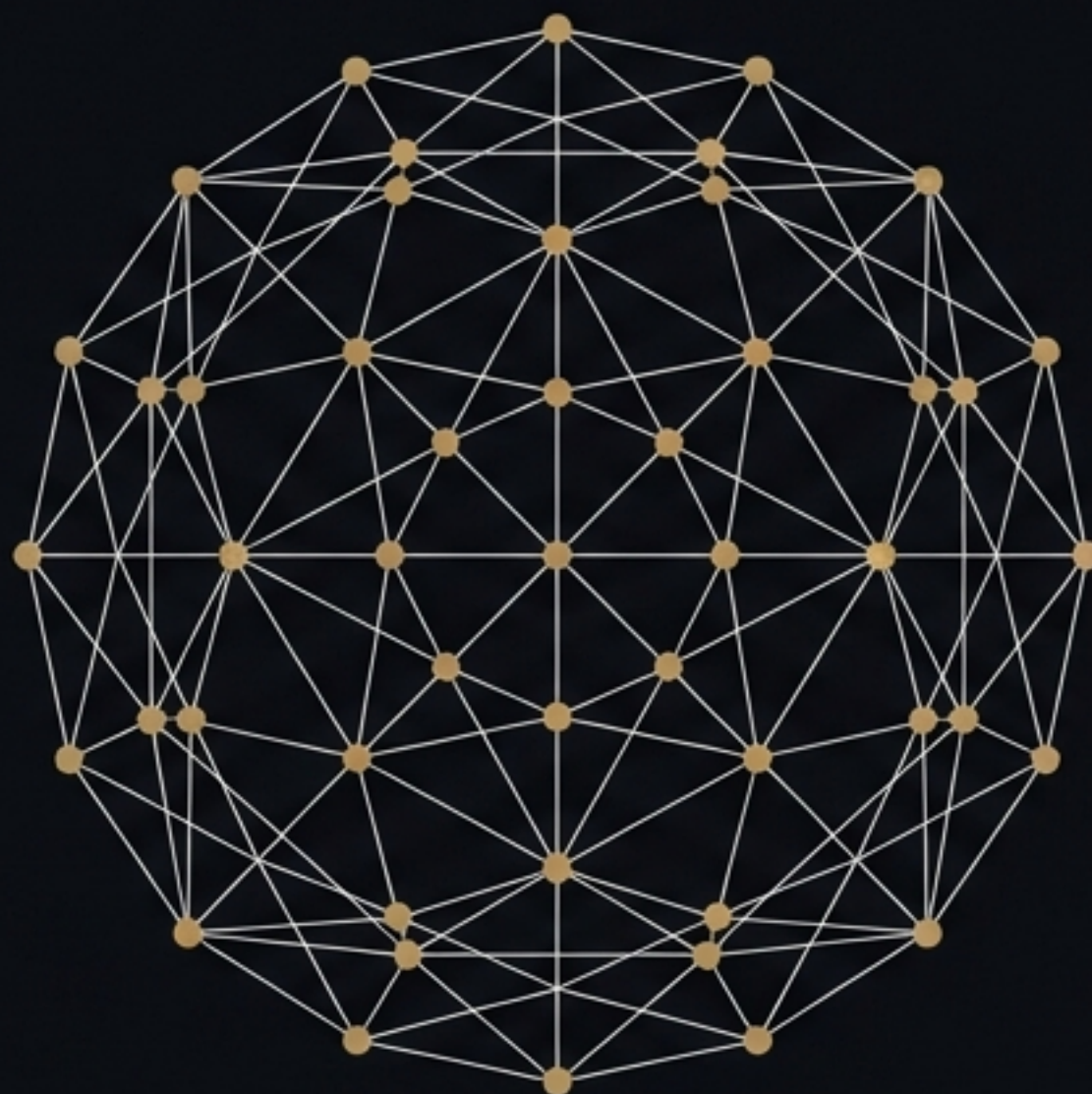


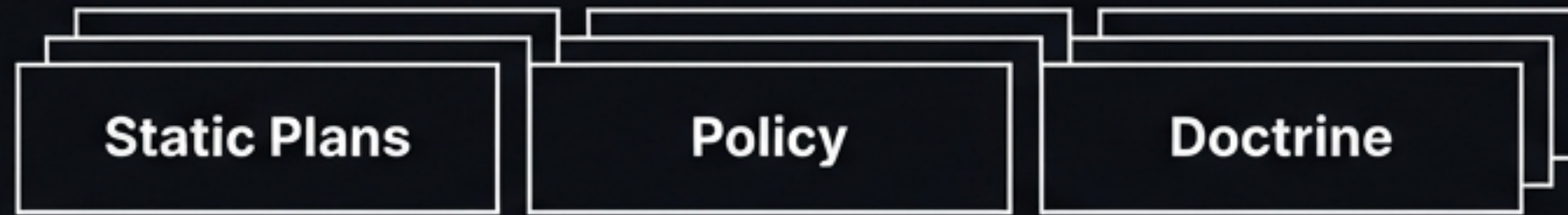
SENTRAIS: TECHNICAL ARCHITECTURE & OPERATIONAL GOVERNANCE

Infrastructure, Security, and Orchestration Framework for High-Stakes Environments



The Systems Challenge: Latency Between Planning and Runtime Execution

Planning & Strategy
(The PDF)



0% - Predictive capability in legacy systems

THE INTELLIGENCE GAP

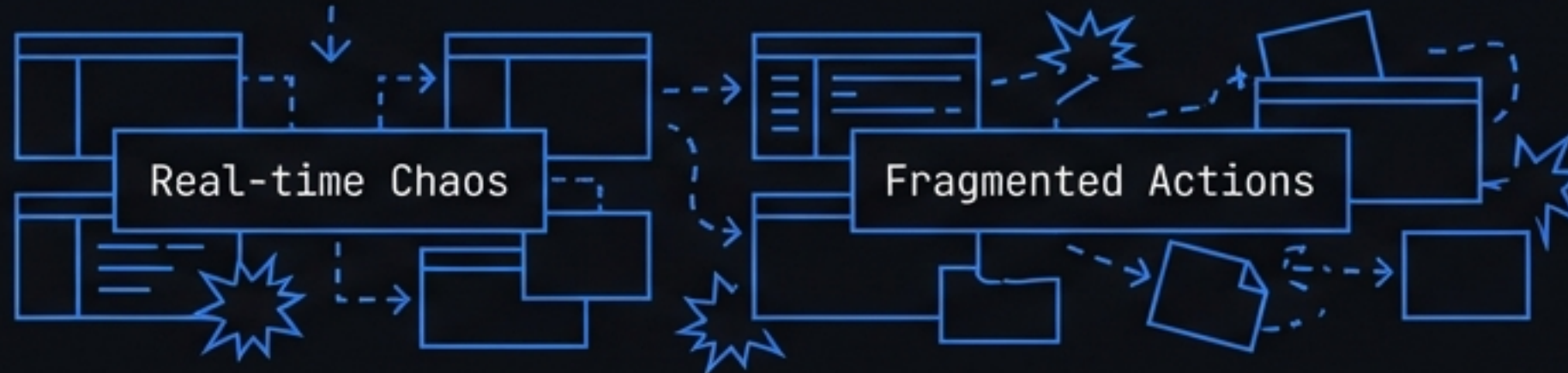
Siloed Spreadsheets

Manual Phone Calls

Disconnected Data

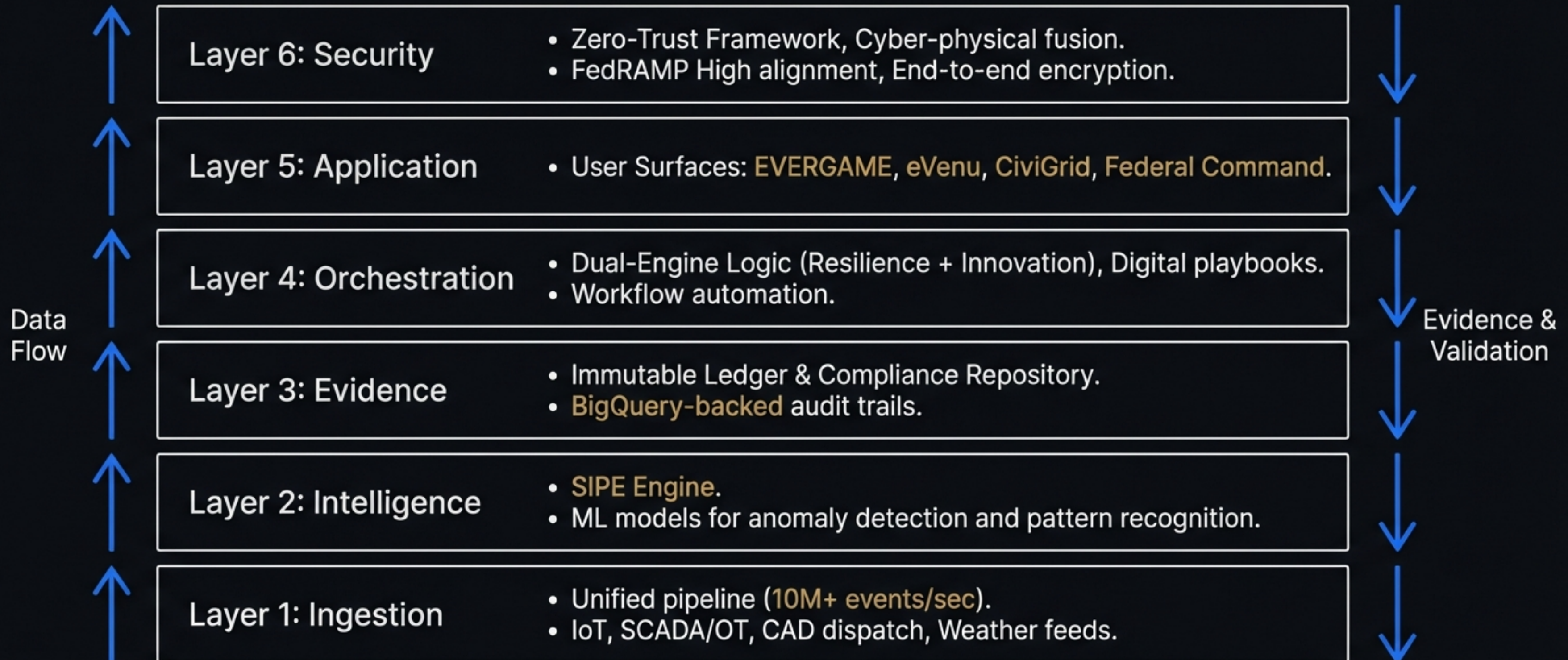
40+ - Agencies per SEAR event operating with no shared picture

Execution & Runtime
(The Crisis)

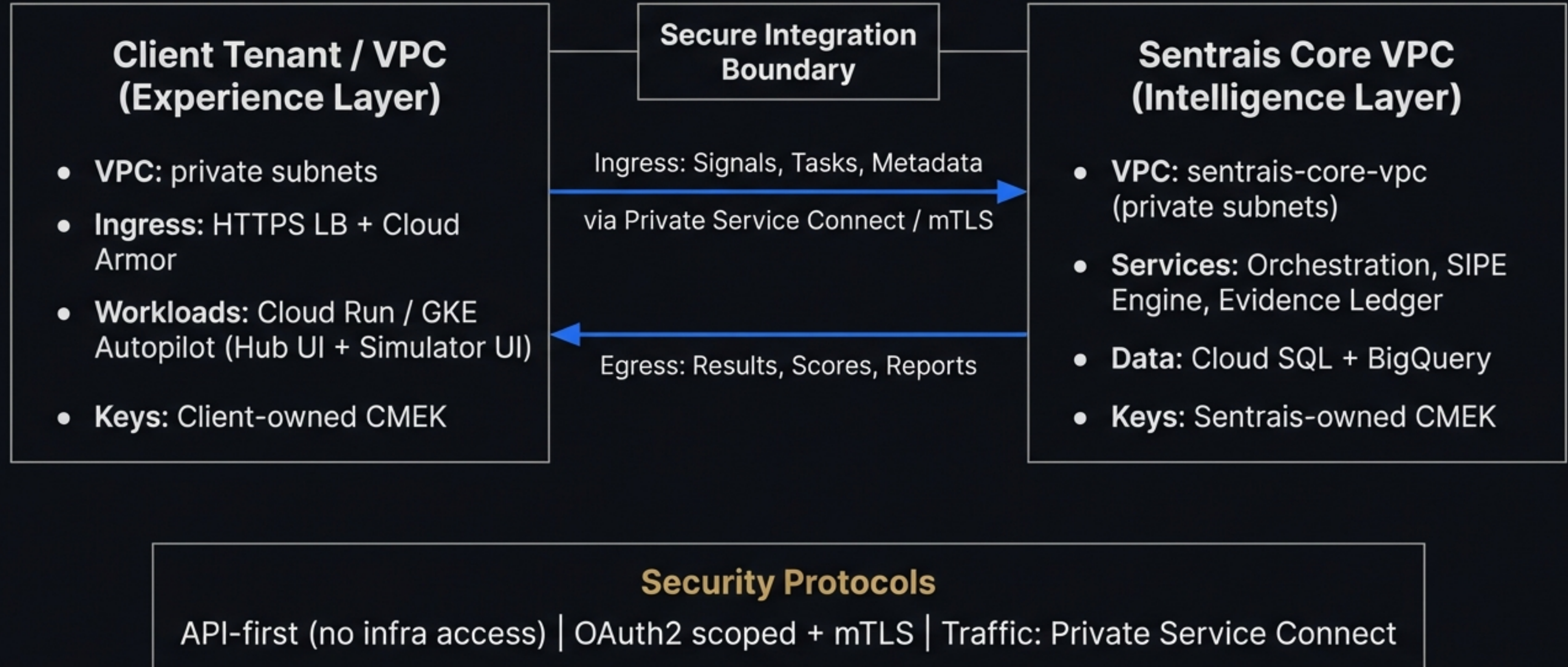


\$51M - Cost of Atlanta Ransomware Attack (Coordination Failure)

High-Level Architecture: The Sentrais Six-Layer Intelligence Stack

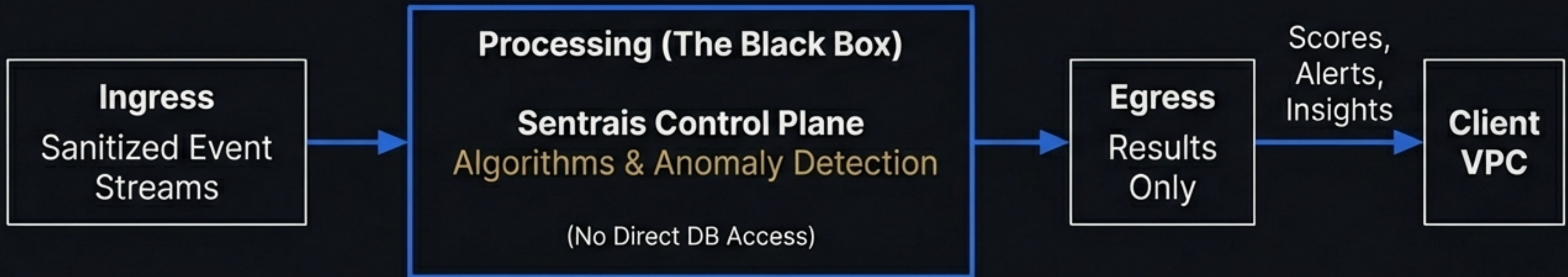


Network Architecture: The Dual-Tenant Delivery Model



Data Flow, Sovereignty, and The Integration Boundary

Client Retains (Sovereign Data)	Sentrai Retains (IP & Logic)
• Operational Data & Logs • User PII & Identity • Brand Assets & Content • Game Day Event Streams	• Source Code & Binaries • Model Weights & Training Data • NIN Methodology Frameworks • Evidence Ledger Schema



Security Note: Data is processed, not co-mingled. The separation is a risk control.

The 'Black Box' IP Framework as a Security Control

Text Concept

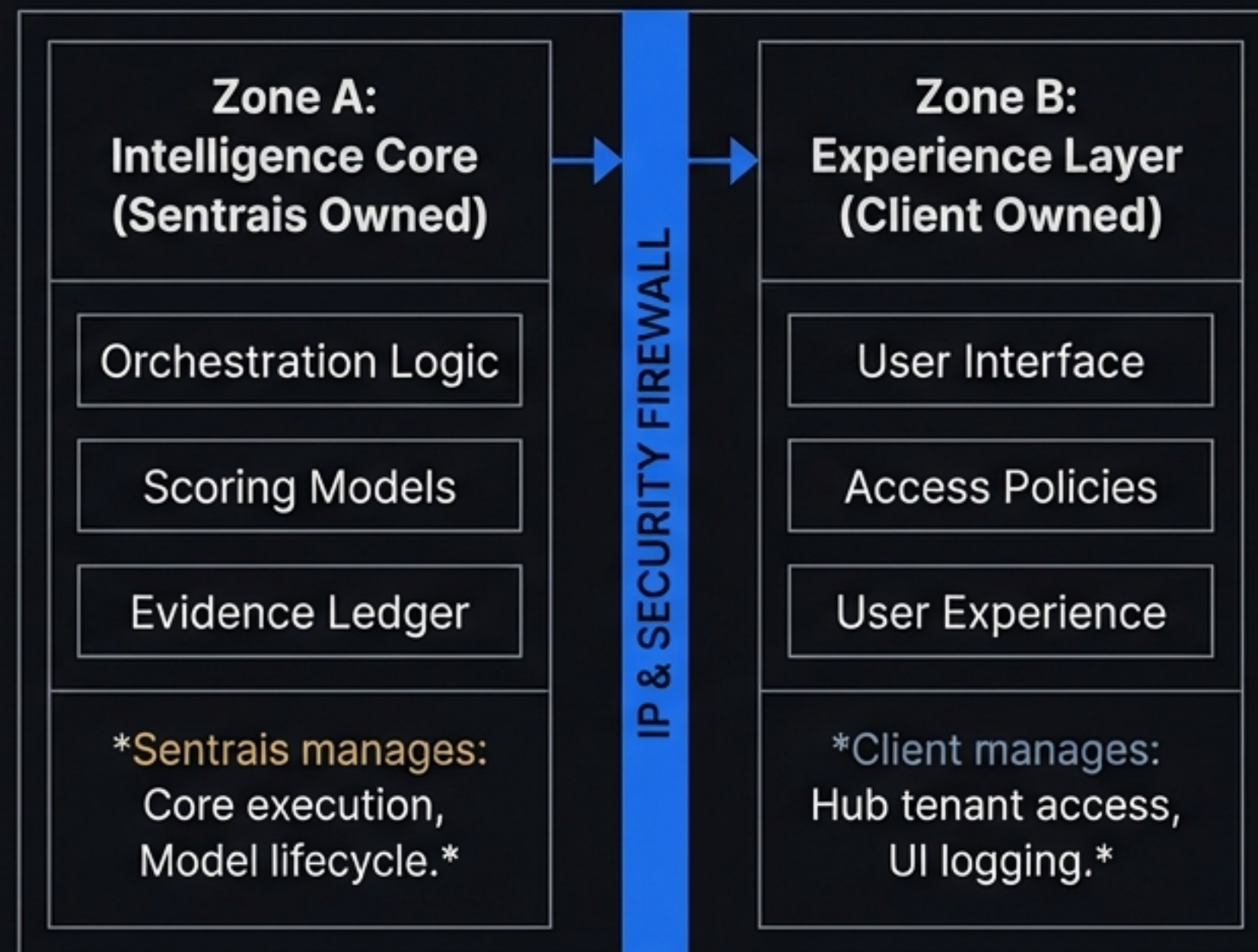
Concept:

Structural separation ensures the “**Secret Sauce**” (**Orchestration Logic**) is never exposed to attack surfaces in the client environment.

Governance Principle:

Control follows responsibility. Ownership follows control. Access does not imply rights.

IP Boundaries Matrix



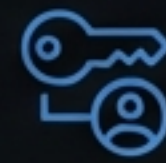
Integration Standards & Security Protocols



Encryption

At Rest: AWS KMS / Cloud KMS (Managed Keys)

In Transit: TLS 1.3 (Minimum for all communications)

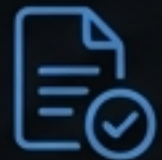


Authentication & Access

Service-to-Service: OAuth2 Scoped Tokens + Mutual TLS (mTLS)

Human Access: MFA Required (Administrative Only)

Session Limits: 4-Hour Maximum Duration



Compliance Alignment

SOC 2 Type II

NIST 800-53

ISO 27001

FedRAMP High Pathway Alignment



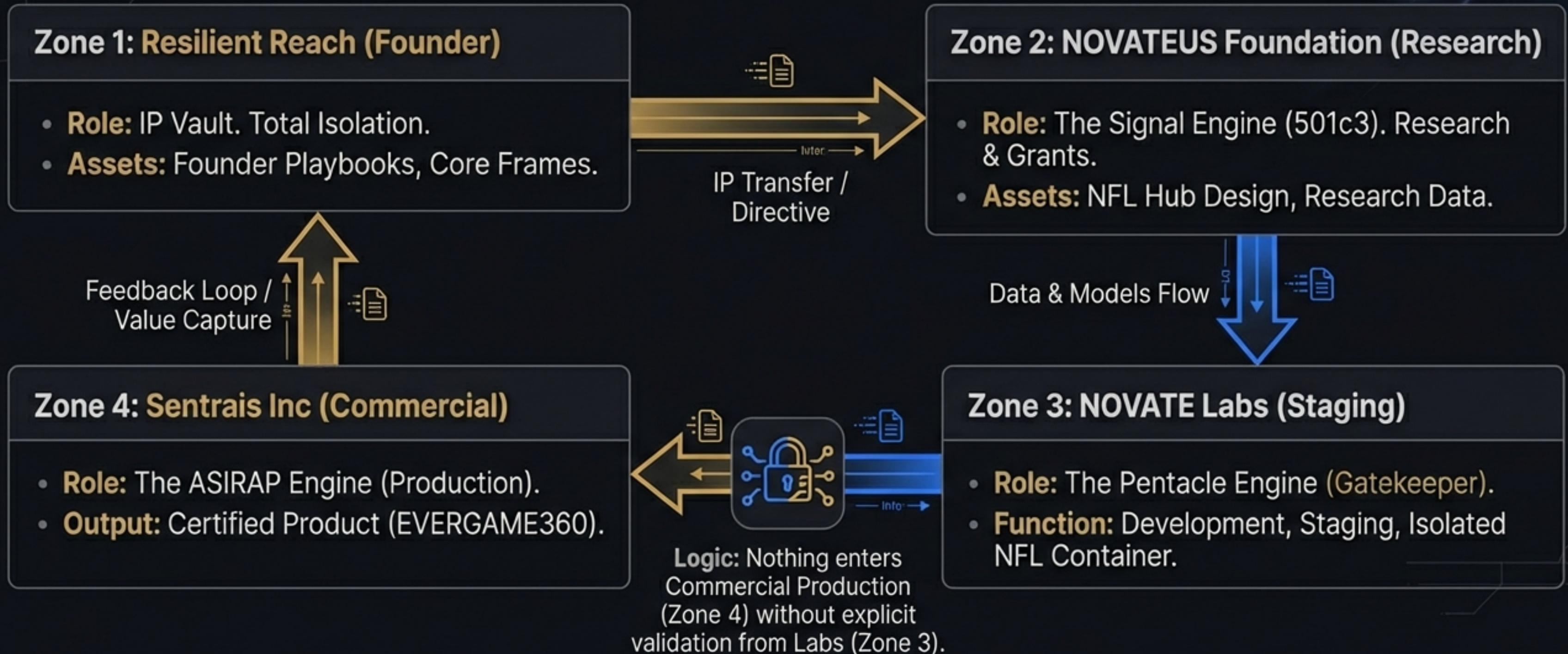
Resilience & Recovery

RTO (Recovery Time Objective): 4 Hours

RPO (Recovery Point Objective): 15 Minutes

Architecture: Cross-Region Replication

CHIMERA: The Four-Zone Governance Architecture



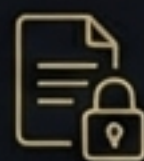
The Evidence Ledger: Immutable Operational Audit

Immutable Event Log (BigQuery Backed)



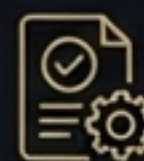
Immutable Logging

Tracks decisions, not just data. Every system state is recorded.



Compliance Automation

Auto-generates After-Action Reports (AAR) for FEMA/Regulatory audits.

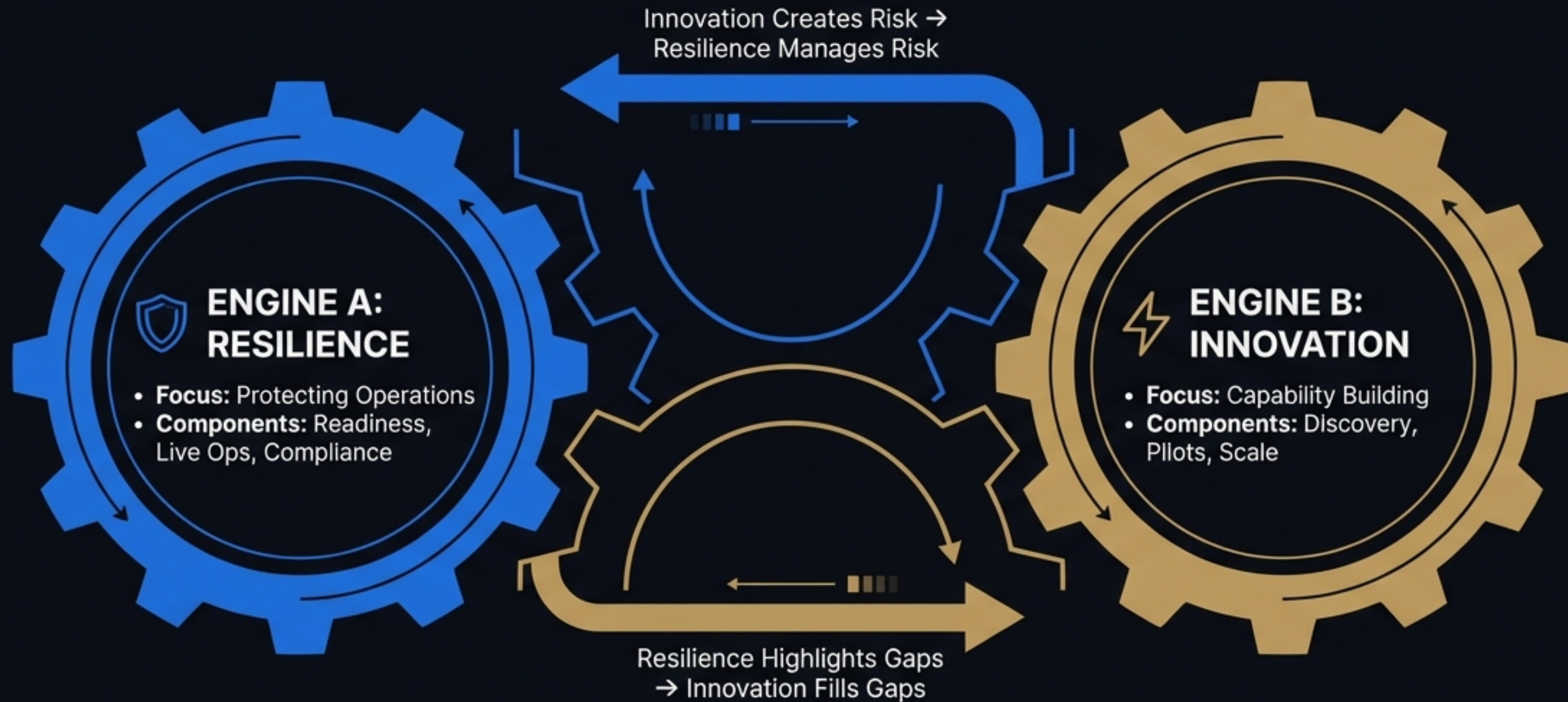


Liability Reduction

Forensic audit trail proves 'Duty of Care' was met during incidents.



Dual-Engine Logic: Synchronizing Resilience & Innovation



Strategic Outcome: Unified operations where emergency gaps drive technology priorities, and new tech is rigorously tested against safety protocols.

AI Safety & The Pentacle Engine Governance

Zone 3 Governance Layer



Calm Design Philosophy: The system orchestrates information for the commander; it does not replace the command chain. Human-in-the-loop is mandatory for critical decisions.

Deployment Methodology: NIN Forensics



Sentrais NIN Delivery: 3-5 Months

→ Outcomes in months, not years.

Traditional Enterprise Tech: 18-24 Months

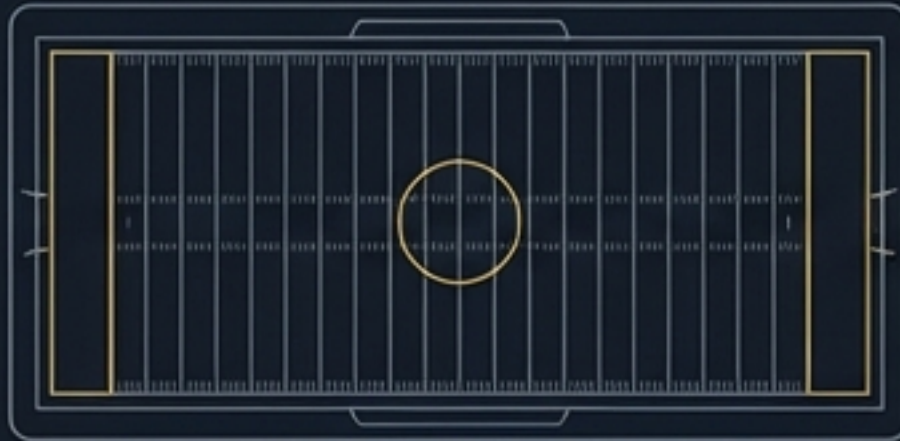
The Annual Certification Dependency Model



Consequence of Non-Renewal: Loss of real-time intelligence, vendor certification status, and predictive analytics.

Validation at Scale: Operations That Cannot Fail

Case Study: NFL (EVERGAME360)



Scale:
32 Clubs
272 Games
100% Adoption



Performance:

96.8

Coordination
Score



Response:

2.3 min

Avg Response
Time



Impact:

\$3.2M

Annual Liability
Reduction

Case Study: NCICC / FEMA

Mission:

Cyber-Physical Fusion
(Crisis Response)

Capability:

Lifeline Modeling
(Energy, Water, Transport)



Strategic Implication:

Architecture for World Cup 2026 &
LA Olympics 2028

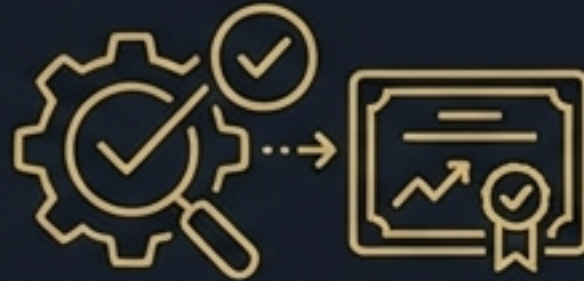
Strategic Technical Summary

Dual-Tenant Security



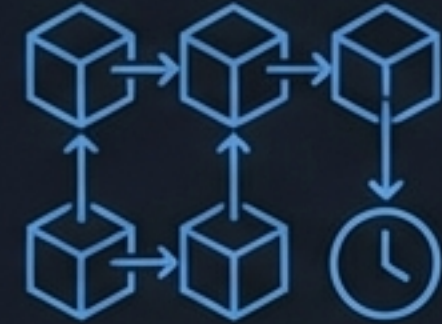
Client owns the experience;
Sentrais protects the
intelligence.

Certified Readiness



Operations moved from
'Hope' to 'Evidence.'

Immutable Trust



A ledger that ensures
compliance and drives
continuous learning.

JOIN THE ORCHESTRATION.

Sentrais Technical Leadership & Architecture Team
confidential@sentrais.com